FITBUDDY

AI Fitness Plan Generator Using Gemini Models

PROJECT REPORT

1. Project Description

FitBuddy is a web-based AI fitness assistant that generates personalized workout plans and nutrition or recovery tips using Google's Gemini models. The application is designed for users with different fitness goals and intensity preferences.

- Generates a personalized 7-day workout plan based on user details, fitness goal, and preferred intensity.
- Provides a concise nutrition or recovery tip related to the user's fitness objective.
- Allows users to submit feedback and dynamically refine their workout plan.
- Provides an administrative view of registered users and their workout plans.

2. Objectives

- To develop an AI-powered fitness planning application.
- To generate structured and personalized weekly workout plans.
- To provide practical nutrition and recovery guidance.
- To support feedback-based modification of workout plans.
- To store user and plan information for future reference.
- To provide a simple interface suitable for users and coaches.

3. Scenarios

Scenario 1 – Personalized Workout Generation

The user enters their name, age, weight, fitness goal, and preferred workout intensity. FitBuddy uses Gemini to generate a structured 7-day plan containing warm-ups, main workouts, and cooldown or recovery guidance.

Scenario 2 – Feedback-Based Plan Refinement

The user can submit feedback such as requesting more cardio or additional rest days. The application sends the existing plan and feedback to Gemini and generates a revised plan.

Scenario 3 – Nutrition and Recovery Tips

FitBuddy generates a concise nutrition or recovery recommendation based on the user's selected fitness goal.

Scenario 4 – Admin Dashboard

An administrator or coach can view registered users and their original and updated workout plans for basic monitoring and comparison.

4. System Architecture

FitBuddy follows a modular architecture in which the browser communicates with a FastAPI backend. The backend handles validation, business logic, AI requests, template rendering, and database operations.

Layer	Technology	Purpose
Frontend	HTML, CSS, Jinja2	User input forms and presentation of workout plans and tips
Backend	FastAPI	Routing, validation, business logic, and API handling
AI Layer	Google Gemini	Workout generation, plan refinement, and nutrition/recovery tips
Database	SQLite + SQLAlchemy	Persistent storage of users and workout plans
Server	Uvicorn	Runs the FastAPI application

Application Flow:

- The user opens the application and enters the required fitness information.
- FastAPI validates the submitted information and sends appropriate prompts to Gemini.
- Gemini generates the workout plan and nutrition or recovery tip.
- The generated results are displayed to the user and stored in SQLite.
- When feedback is submitted, the existing plan is refined and the updated version is stored.
- The administrator can view stored users and plan information through the dashboard.

5. Technologies and Requirements

Technology / Requirement	Purpose
Python 3.10+	Application development and execution
FastAPI	Backend web framework
Uvicorn	ASGI application server
Jinja2	HTML template rendering
SQLAlchemy	Database ORM
SQLite	Lightweight persistent database
Google Gemini API	AI-powered plan and tip generation
python-dotenv	Environment variable and API-key management
HTML & CSS	Frontend development
Git	Version control

6. Development Milestones

Milestone 1 – Model Selection and Architecture

The AI layer required a model capable of generating structured workout plans, responding to user feedback, and producing concise wellness guidance. Gemini models were selected for their API-based integration and ability to handle structured natural-language prompts.

Model	Purpose
Gemini 1.5 Pro	Workout plan generation and feedback-based plan refinement
Gemini Flash	Fast nutrition and recovery tips

Milestone 2 – Core Functionality

- Workout Plan Generation: Creates a structured 7-day plan based on user inputs.
- Nutrition Tip Generation: Produces a concise recommendation related to the user's goal.
- Feedback-Based Plan Update: Revises the existing plan using user feedback.
- User and Plan Storage: Saves user details and preserves original and updated plans.
- Administrative View: Displays users and their available plan information.

Milestone 3 – Backend Development

Route	Purpose
/	Displays the home page and input form
/generate-workout	Processes user input and generates the plan
/submit-feedback	Updates an existing plan using user feedback
/view-all-users	Displays the administrative user and plan view

Milestone 4 – Frontend Development

Template	Purpose
index.html	Collects user information
result.html	Displays the workout plan, tip, and feedback form
all_users.html	Displays users and their workout plans

- Responsive form and result layouts.
- Clear input fields and action buttons.
- Readable workout-plan presentation.
- Consistent styling using CSS and web fonts.
- Jinja2 template variables for dynamic content.

Milestone 5 – Local Deployment

- Create and activate a Python virtual environment.
- Install project dependencies.
- Store the Gemini API key securely in environment variables.
- Start the application using Uvicorn.
- Verify the main application, administrative page, and API documentation.

7. Functional Testing

Test No.	Test Case	Expected Result
1	Open the home page	Input form loads correctly
2	Submit user details	A personalized 7-day plan and nutrition/recovery tip are generated
3	Submit feedback	The workout plan is revised according to the feedback
4	Add another user	User records remain separate and are stored correctly
5	Open the admin page	Registered users and plan information are displayed
6	Open API documentation	Available API operations are displayed and testable

8. Expected Outcomes

- A functional web application for AI-assisted fitness planning.
- Personalized weekly workout plans based on user inputs.
- Goal-oriented nutrition and recovery guidance.
- Dynamic plan refinement through user feedback.
- Persistent storage of user and workout-plan information.
- A basic administrative interface for monitoring users and plans.

9. Conclusion

FitBuddy demonstrates the integration of generative AI into a practical fitness application. By combining FastAPI, Jinja2, SQLAlchemy, SQLite, and Google Gemini models, the system can generate personalized workout plans, provide nutrition or recovery guidance, and refine plans based on user feedback. The modular design also provides a foundation for future improvements such as progress tracking, mobile support, multilingual interaction, wearable integration, and richer AI-assisted fitness features.

10. Future Enhancements

- Voice-based interaction and multilingual support.
- Mobile application support with suitable offline features.
- Progress tracking and fitness dashboards.
- Gamification and achievement features.
- Integration with wearable fitness devices.
- PDF and image-based fitness-plan input using multimodal AI capabilities.

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